Seminar 08 cheat sheet

Constrained Optimality and Discrete Modelling

Optimization Methods
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Course lecturer: Andrey Ignatov
Author: Evgeny Baulin, Telegram @tarakan_tuc

KKT conditions

- $L = f + \sum \lambda_i g_i + \sum \nu_j h_j$, $g_i \le 0$, $\lambda_i \ge 0$, ν_j free.
- Stationarity $\nabla_x L = 0$, feasibility, $\lambda \ge 0$, $\lambda_i g_i = 0$.
- Necessary under LICQ (or, for a convex problem, a strictly feasible point); sufficient when the problem is convex.
- Slack $\Rightarrow \lambda_i = 0$; tight does not imply $\lambda_i > 0$.

Active-set recipe

- Pick A: $g_i = 0$ on A, $\lambda_i = 0$ off A; solve.
- Reject if some $g_i > 0$ or $\lambda_i < 0$.
- Convex: the survivor is global. Nonconvex: compare all survivors.

Duality and prices

- $q = \inf_x L$ is concave; $q(\lambda, \nu) \le f(x)$ for feasible x, $\lambda \ge 0$.
- Convex + strictly feasible point: $d^* = p^*$, dual optimum attained.
- $g_i \le b_i$: $\partial p^* / \partial b_i = -\lambda_i^*$. For $\ge$ the sign flips.
- $f(x) - q(\lambda) \ge 0$ certifies how far a feasible x can be from optimal.

Integer models

- Decisions → variables, rules → constraints, goal → objective.
- Relaxation: max gives an upper bound, min a lower bound; rounding is not a method.
- $x \le Mz$ switches a fixed cost; $z_A \le z_B$ is if A then B; $\sum z_i \le k$ caps a choice.
- Take M as small as valid: same integer solutions, tighter relaxation.

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